

ARI 2129 - Principles of Computer Vision for AI

Adversarial Quiz Rationale: Otsu's Thresholding

<https://forms.gle/Mcm3quvenvyHWHAA6>

This document provides the rationale for the Adversarial Quiz, explaining the logic behind the correct answers and the nature of the distractors (misconceptions).

Q1 – Objective of Otsu

Correct Answer: B. Otsu minimises within-class variance.

Distractor Rationale:

A reflects a common misconception that thresholding improves brightness.

C confuses thresholding with edge detection.

D assumes thresholding removes noise, which is incorrect.

Q2 – Histogram Shape

Correct Answer: B. Works best for bimodal distributions.

Distractor Rationale:

A and C ignore the need for separable classes.

D reflects the misconception that skewed data is sufficient.

Q3 – Unimodal Failure

Correct Answer: B. No clear class separation.

Distractor Rationale:

A and C are unrelated.

D confuses colour diversity with histogram structure.

Q4 – Key Assumption

Correct Answer: B. Two-class separation.

Distractor Rationale:

A is secondary, not core.

C assumes a noise model unnecessarily.

D is irrelevant.

Q5 – Uneven Illumination

Correct Answer: B. Global threshold fails.

Distractor Rationale:

A reverses expected behaviour.
C and D are unrelated effects.

Q6 – Alternative Method

Correct Answer: B. Adaptive thresholding.

Distractor Rationale:

A and C are preprocessing, not solutions.
D is unrelated.

Q7 – Global Thresholding

Correct Answer: B. One threshold for entire image.

Distractor Rationale:

A and C describe adaptive methods.
D is incorrect.

Q8 – Variance Formulation

Correct Answer: B. Maximises between-class variance.

Distractor Rationale:

A, C, D misunderstand optimisation objective.

Q9 – Noise Effect

Correct Answer: C. Distorts histogram.

Distractor Rationale:

A ignores impact.
B and D reflect incorrect assumptions.

Q10 – Failure Case

Correct Answer: C. Multiple intensity regions.

Distractor Rationale:

A and B are ideal conditions.
D is misleading.

Overall, this quiz challenges intuitive but incorrect assumptions, ensuring that learners demonstrate true conceptual understanding rather than memorisation.